Duy Vu

Machine Learning Engineer

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SUMMARY

Machine Learning Engineer at Nokia with 5+ years building end-to-end ML pipelines for deep learning-based 5G/6G receivers: data generation, training, quantization, evaluation, and interpretability. Also experienced in LLM evaluation for RAG chatbots. MSc in Machine Learning, Data Science and AI (Aalto University).

WORK EXPERIENCE

Machine Learning Engineer

Nokia, Espoo, Finland – MN RAN L1 Software | *May 2021 – Present (joined as ML Trainee)*

- Develop, maintain, and run the end-to-end ML pipeline for deep learning-based 5G/6G physical-layer receivers
- Generate, verify, and process training data; perform exploratory data analysis
- Train, fine-tune, quantize, and evaluate models
- Plan and coordinate model training workflow development with cross-functional teams
- Design and build interpretability tools for the deep learning receiver, including Neural Activation Pattern (NAP)-based concept discovery and out-of-distribution (OOD) detection
- Tech: PyTorch, Captum, Plotly, Matplotlib, Amazon S3, MLflow, Kubeflow, Docker, Gerrit, Grafana, Linux, PlantUML

ML Engineer – LLM Evaluation (Volunteer)

Tanzania AI Community × AaltoAI – Twiga | *April 2026 – July 2026*

- Built an automated evaluation system for Twiga, a RAG-based WhatsApp chatbot for Tanzanian teachers (Meta Llama Impact Grant awardee)
- Co-curated a 500-question evaluation dataset across school subjects, generating questions from textbook chunks with LLMs and hand-reviewing them for quality
- Implemented RAG evaluation metrics: faithfulness, answer relevancy, factual correctness, LLM-as-a-judge, and lexical matching (ROUGE)
- Benchmarked 2 production LLMs on the dataset and surfaced results in a Streamlit dashboard
- Tech: Python, Ragas, DeepEval, LangSmith, OpenAI API, Llama, Streamlit, pandas, pytest

Teaching Assistant

Aalto University, Finland | *2023*

- Course: Information Visualization
- Supported students with lecture content and assignments
- Graded assignments and exams

Teaching Assistant

Tampere University, Finland | *2020 – 2022*

- Courses: Introduction to Machine Learning and Pattern Recognition, Computer Vision
- Supported students with lecture content and assignments
- Graded assignments and exams

Research Assistant

Tampere University, Finland | *2020 – 2021*

- CIVIT Lab: Built a pipeline to extract gaze data (Pupil Core eye tracking), run object detection (Detectron2), and analyze mother-to-infant gaze; extended into BSc thesis (grade 5/5)
- Cloud and Web Engineering group: Mined Apache's GitHub repositories and developed a tool to detect refactorings in repository histories using RefactoringMiner

EDUCATION

Master of Science (Technology)

Aalto University, Finland | *September 2022 – June 2025*

- Major in Machine Learning, Data Science and Artificial Intelligence

- Thesis: Learned layer-level concepts in the machine learning model on the physical layer in 6G L1 (explainable AI, clustering, concept discovery)
- GPA: 4.75/5.0

Exchange Study
Karlsruhe Institute of Technology (KIT), Germany | October 2023 – March 2024
- Economics and Management (Wirtschaftswissenschaften)

Bachelor of Science and Engineering
Tampere University, Finland | September 2019 – May 2022
- Major: Information and Communication Technology
- Minor: Mathematics
- Second minor: Signal Processing and Machine Learning
- GPA: 4.79/5.0

PUBLICATIONS

- Tuononen, M., Penttinen, H., Vu, D., Korpi, D., Starck, V., & Hautamäki, V. (2025). Out-of-Distribution Detection via Channelwise Feature Aggregation in Neural Network-Based Receivers. arXiv preprint, arXiv:2505.15570.

PROJECTS

Bird Audio Detection Kaggle Challenge
- Use different preprocessing methods and fine-tune a Convolutional Recurrent Neural Network to detect bird sound in audio
- Top 5 in private leaderboard, top 11 on public leaderboard, out of over 70 teams
- Tools: Tensorflow, librosa

Acoustic Scene Classification with Multiple Devices (DCASE 2020)
- Inspired by top submissions: transform audio to log-mel spectrogram format, compute deltas & delta-deltas features
- Implement a ResNet fusion model learning features from different frequency-bin segments and merging them to predict the target class
- Without data augmentation, the model achieves ~77% accuracy on the development test dataset (top 5)

AI-Based Demand Forecasting for Porsche AG
- Developed the model pipeline for data preprocessing, training, and testing
- Data analysis
- Build time series models: ARIMAX, XGBoost, Prophet, and a metamodel of all three

SKILLS

- **Programming Languages:** Python, MATLAB, C++, R, Scala, JavaScript, SQL
- **Libraries:** PyTorch, TensorFlow, scikit-learn, Captum, pandas, NumPy, Matplotlib, Plotly, Streamlit
- **LLM & Generative AI:** LLM Evaluation, Retrieval-Augmented Generation (RAG), LLM-as-a-Judge, Ragas, DeepEval, LangSmith, OpenAI API
- **Database:** MongoDB
- **Other Technologies:** Docker, Kubernetes, Git, Gerrit, MLflow, Kubeflow, Apache Spark, Amazon S3, Linux/Unix
- **Industry Knowledge:** Agile Methodology, Software Design
- **Backend Development:** Flask
- **Languages:** Vietnamese (Native), English (Professional working proficiency), Finnish (Elementary proficiency)

CERTIFICATIONS

- Deep Learning Specialization – DeepLearning.AI
- Business Intelligence Capstone Project – Karlsruhe Institute of Technology (KIT)